A Project Report on

Nirbhaya Path

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

This is to certify that the project entitled "**Nirbhaya Path**" is a bonafide work of **Arihant Chipper (24106179), Nishit Jain (24106141), Yashwant Injapuri (24106093)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

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Project Report Approval for S.E.

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Abstract

In today's increasingly unpredictable environment, personal safety has become a paramount concern, yet many individuals still rely on traditional emergency methods like manual dialing or messaging during critical situations. These methods are often too slow, conspicuous, or physically impossible to use when a person is under immediate threat. To address these challenges, the present study proposes Nirbhaya Path, a specialized, user-centric web application designed for rapid and discreet emergency response on mobile devices. The system features a unique "Hold-to-Activate" SOS trigger that prevents accidental alerts while ensuring a fast 3-second response time, alongside a Fake Call utility that uses realistic system-style overlays to provide users with a safe "social exit" from uncomfortable situations. Experiments and user-testing were performed to evaluate the reliability of the real-time geolocation tracking and haptic feedback mechanisms. The results demonstrate that the proposed method is a feasible and scalable solution for personal protection in high-stress environments. This work contributes to SDG 9 (Industry, Innovation and Infrastructure) by promoting advanced, lightweight technological solutions, SDG 11 (Sustainable Cities and Communities) by enhancing personal safety in urban environments, SDG 3 (Good Health and Well-being) by reducing physical and psychological risk, and SDG 5 (Gender Equality) by providing tools to deter harassment and ensure safer public participation. The research demonstrates the potential of Progressive Web App (PWA) technologies in facilitating a scalable and sustainable digital shield for personal protection.

Keywords: Personal Safety, SOS Trigger, Fake Call Utility, Real-time Tracking, Progressive Web App (PWA)

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ABBREVATION

AI: *Artificial Intelligence*

ML: *Machine Learning*

PWA: *Progressive Web App*

SOS: *Save Our Souls (Standard emergency distress signal)*

GPS: *Global Positioning System*

API: *Application Programming Interface*

UI: *User Interface*

UX: *User Experience*

HTML5: *HyperText Markup Language version 5*

CSS3: *Cascading Style Sheets version 3*

RWA: *Responsive Web Application*

IoT: *Internet of Things*

NoSQL: *Not Only SQL (referring to the database type)*

IDE: *Integrated Development Environment*

MQTT: *Message Queuing Telemetry Transport*

Chapter 1

Introduction

1.Introduction

The rapid advancement of digital technologies has fundamentally transformed how individuals manage their personal safety and emergency responses. Mobile applications and web-based services have made real-time tracking and distress signaling more accessible than ever before. However, despite these advancements, many people still rely on traditional, high-friction methods during emergencies, such as manually dialing emergency numbers or sending text messages. These methods are often time-consuming, physically difficult to perform under stress, and highly conspicuous.

One of the major challenges in personal safety is the difficulty of triggering a distress signal discreetly and rapidly when faced with a perceived threat. In many high-pressure situations, victims are unable to navigate complex smartphone interfaces or wait for applications to load, which can waste critical seconds. Furthermore, a lack of real-time location sharing and proactive deterrence features, such as simulated social "exits," further complicates the safety process. Guardian SOS is introduced as a comprehensive personal safety management system designed to simplify and improve the process of emergency alerting. Guardian SOS allows users to trigger a silent distress signal through a 3-second hold mechanism and provides a "Fake Call" utility to help users safely exit

uncomfortable social situations. The system focuses on improving response speed, ensuring user discretion, and enhancing interaction between the user and their emergency contacts.

1.1 Background and Motivation:

The motivation behind this project arises from the increasing demand for efficient and accessible personal safety tools in urban environments. In a fast-paced world, individuals expect safety solutions that are immediate and require minimal cognitive effort during a crisis. Traditional systems often fail to meet these expectations due to procedural delays and a lack of discreet activation methods.

While existing safety platforms have attempted to bridge this gap, many users find them difficult to use due to complex navigation and excessive features that are not helpful in a moment of panic. There is a growing need for a system that provides location-based alerts and social-safety simulations (like fake calls) quickly and easily. The Guardian SOS system is motivated by the need to create a simple, fast, and reliable platform that enhances user confidence and improves accessibility to emergency support.

1.2 Need for the Proposed System:

The need for the proposed system arises due to the limitations of traditional emergency responses and current safety applications. Users require a platform that provides accurate, real- time coordination with their trusted circle.

The key needs of the system are:

- To eliminate dependency on manual, slow emergency dialing methods.
- To provide real-time location updates to emergency contacts.
- To help users exit suspicious social encounters discreetly.
- To reduce the "trigger-to-alert" time during a physical crisis.
- To offer a simple, mobile-optimized interface that works under stress.
- To ensure secure and private management of user contact data.

1.2.1 Overview of Nirbhaya Path System:

To address these identified problems, Guardian SOS is proposed as a mobile-first web application. The system provides a centralized platform for users to manage emergency contacts and trigger rapid alerts. Guardian SOS uses high-precision location services to help contacts find the user's exact coordinates.

The system allows users to customize their safety experience through modules like the "Fake Call" simulation, which includes realistic haptic vibration and caller UI to deter potential harassers. It also provides real-time SOS broadcasting, which helps in reducing response delays. Additionally, the system ensures secure storage of the user's emergency circle and provides automated location sharing during an active event. The main aim of Guardian SOS is to provide a fast, user-friendly safety solution that improves individual protection in public spaces.

1.3 Scope of the Project:

The scope of the Guardian SOS system includes the development of a responsive safety platform that simplifies distress signaling and social deterrence.

The system focuses on:

- Rapid SOS triggering through a hold-to-activate mechanism.
- Generating convincing "Fake Call" simulations with haptic feedback.
- Managing and synchronized emergency contact circles.
- Providing real-time GPS location broadcasting.
- Ensuring data privacy for user location and contact information.

Chapter 2

Literature Survey/ Existing system

2. Literature Survey

The development of personal safety applications has seen a significant surge in recent years, driven by the increasing need for rapid emergency response and discreet deterrence mechanisms. Researchers have explored various technological domains, including IoT-based tracking, mobile sensor fusion, and psychological deterrents like fake-call interfaces. This chapter provides a detailed review of existing research and systems that have contributed to the evolution of the **Guardian SOS** project.

[1] "IoT-Based Emergency Alert and GPS Tracking System" (2025) The authors, in this IEEE-indexed study, developed a hardware-interfaced system using ESP32 and A9G modules. Their research highlights the reliability of MQTT protocols for real-time location broadcasting in low-connectivity scenarios. While highly effective, the study notes that hardware-dependent systems are often bulky and less accessible to the general public compared to software-only mobile solutions.

[2] "Intelligent Emergency Detection using Smartphone Sensor Fusion" (2024) This paper explores the use of accelerometers and gyroscopes to automatically detect falls or high-impact distress. The researchers achieved a 92.3% detection accuracy. However, they identified a significant research gap: automated systems often suffer from high "false-positive" rates during normal activities (like running or dancing), emphasizing the need for a manual "Hold-to- Activate" trigger like the one implemented in Guardian SOS to ensure user intent.

[3] "Discreet Safety Features in Mobile Apps for Sexual Violence Prevention" (2024) A scoping review published in PMC analyzed apps like *uSafeUS* and *Co6*. The study found that "Fake-a-Call" features are among the most valued utilities for users in uncomfortable social situations. The research concludes that providing a "social exit" strategy is just as critical as providing an emergency alert, as it can prevent an escalation of a threat before it becomes a physical emergency.

[4] "Fraudulent and Fake Calls in India: A Comprehensive Review" (2024) While primarily focused on security, this ResearchGate paper discusses the psychological impact of incoming calls. It highlights that the "authority" associated with an incoming call can effectively distract or deter a potential harasser. This finding provided the psychological foundation for our Fake Call module—mimicking the exact UI and haptic patterns of a real call to maximize the deterrent effect.

[5] "Improvement of Emergency Situation Management through Integrated Systems" (2022) Published in MDPI, this study emphasizes that reducing response time by just 1 minute can increase the chances of saving a life by up to 6%. The authors advocate for "one-tap" or "gesture-based" activations. Our project builds on this by implementing a 3-second hold, which is faster than unlocking a phone and navigating to a dialer, yet more secure than a single accidental tap.

[6] "Sensor-Driven Emergency SOS Mobile Applications" (2025) This recent research in IJERD analyzed existing Android apps and found that most lacked "discreet activation methods" that don't require the user to look at the screen. The paper suggests that haptic feedback (vibration) is essential to confirm to the user that an alert has been sent without alerting an attacker.

Existing platforms such as Life360 have gained popularity for providing family location sharing and safety services. Life360 allows users to create "circles" and view real-time

movements of family members. However, it has limitations such as heavy battery consumption due to constant GPS pings and complex navigation that can be overwhelming during a high- stress emergency. Additionally, its primary focus is long-term tracking rather than immediate, discreet distress signaling.

Similarly, Safe provides features like "Follow Me" and emergency recording. While it enhances personal security through voice activation, it often requires a premium subscription for core features and relies heavily on a stable, high-speed internet connection to process voice commands. It is not primarily focused on providing a "social exit" strategy, such as a simulated incoming call, which is often needed in the early stages of a suspicious encounter.

From the above studies, it is clear that while existing systems provide useful features, they still have limitations such as high "trigger-to-alert" latency, poor user experience under panic, heavy dependency on mobile data, and a lack of discreet deterrence mechanisms.

The proposed system, Guardian SOS, aims to overcome these limitations by integrating a rapid 3-second hold-to-activate trigger, a realistic fake call simulation with haptic feedback, and a lightweight mobile-web interface into a single platform. It focuses on providing a simple, efficient, and highly reliable personal safety solution.

2.1 Summary of Literature Findings:

Based on the above literature survey and analysis, the following observations are made:

- **Complexity vs. Speed:** Most systems provide comprehensive safety features but lack the simplicity required for rapid activation during a crisis.
- **Trigger Inefficiency:** Manual dialing or multi-step app navigation is too slow for real- time emergency response.
- **Lack of Discretion:** Existing systems often lack features that allow a user to deter a harasser without escalating the situation (e.g., Fake Call).
- **Hardware & Data Dependency:** Many apps require high-end smartphones or constant 4G/5G connectivity to function effectively.
- **User Interface Complexity:** Overcrowded interfaces increase cognitive load, making the app difficult to use when the user is panicking.

- **Battery & Privacy:** Constant background tracking in existing apps leads to significant battery drain and raises concerns regarding data privacy.

These findings highlight the need for a system that is simple, efficient, discreet, and accessible, which is addressed by the proposed Guardian SOS systems.

Table 1 Literature Survey Summary

Sr.No	Author(s)	Year	Method / Approach	Findings	Limitations
1	Schulze	2024	Scoping Review of Safety Apps [2]	Mobile safety apps significantly reduce harm to at-risk individuals.	Few studies evaluate apps specifically designed for youth/emergencies.
2	B. Durga	2024	AI-based Call Pattern Analysis [3]	Fake calls act as a psychological deterrent by creating a "connected" presence.	Systems struggle to mimic system-level UI realistically for haptics.
3	Ganavi M	2023	IoT and Machine Learning [4]	Real-time sensor data improves reporting speed for emergency authorities.	High false-positive rates when using fully automated detection.
4	Benoit et	2022	Environmental Scan of Safety Tools	One-tap and "low-friction" UI designs are crucial for high-stress usability.	Most existing tools require complex setup or native app installation.
5	**Sharma & Kumar**	2022	GPS and MQTT Protocol [1]	Lightweight web-based protocols ensure reliable location broadcasting.	Network latency can affect real-time updates in low-data zones.

6	Mehta & Shah	2021	UI/UX Cognitive Analysis [5]	Minimalist design reduces user panic and cognitive load during a crisis.	Existing apps often have cluttered interfaces with too many menus.

Table 1: Summarizes the key findings and limitations of the existing systems discussed in the literature survey

Chapter 3

Limitation of Existing system

3. Limitation of Existing system

The existing personal safety applications and digital emergency platforms have significantly improved access to location-based tracking and distress signaling. However, despite these advancements, several limitations still exist, which affect their efficiency, usability, and overall effectiveness during real-world crises.

One of the major limitations is the lack of proactive deterrence mechanisms. Most existing systems are reactive, meaning they only function once a crime or accident has already occurred. They do not provide tools to help users exit uncomfortable social situations discreetly before they escalate. This makes it difficult for users to prevent a threat from turning into a physical emergency.

Another important issue is the complex user interface of many applications. Some platforms are overcrowded with features and menus that are not designed with simplicity in mind. In a high-stress situation, navigating through multiple screens to find an SOS button is difficult,

especially for users experiencing panic. This significantly reduces the overall usability of the system when seconds matter most.

Existing systems also face challenges in providing low-friction activation. In many cases, triggering an alert requires unlocking the phone and finding an icon, leading to fatal delays. This results in longer response times from emergency contacts and increased risk for the user.

Limited accessibility and resource dependency is another major drawback. Many safety applications are heavy native apps that require significant storage and constant background data, which is not always available on entry-level devices or in low-network areas. This restricts access to safety tools for a large portion of the population who may not want to install permanent tracking software.

Data privacy and battery drain are also significant issues in current systems. Continuous GPS tracking in family-circle apps leads to rapid battery exhaustion, often resulting in the phone dying when it is needed most. Additionally, handling sensitive live-location data requires strong security, which not all platforms ensure.

Additionally, many systems do not provide a fast and silent feedback loop. Users often do not know if an alert was successfully sent without looking at their screen, which can be dangerous if they are trying to hide the fact that they are calling for help.

Furthermore, there is a lack of integration of essential haptic features such as realistic vibration patterns for simulated calls. This limits the overall effectiveness of the platform in providing a convincing "social exit" strategy.

These limitations highlight the need for an improved system that focuses on simplicity, rapid hold-to-activate functionality, discreet deterrence, and lightweight web-based access.

3.1 Research Gap :

Based on the analysis of existing systems, the following research gaps are identified:

- **Lack of simple and user-friendly panic interfaces** for high-stress situations.
- **Inefficient trigger mechanisms** that require too many steps to activate.

- **Absence of proactive deterrence tools** like realistic "Fake Call" simultions.
- **High battery consumption** due to non-optimized background tracking.
- **Limited cross-platform accessibility** for users without native app installations.
- **Inadequate tactile feedback** (haptics) to confirm alerts silently.

The proposed system, Guardian SOS, aims to address these gaps by providing a centralized, efficient, and user-friendly personal safety management system that prioritizes speed and discretion.

Table 2 Limitations of Existing Safety Systems

Sr.No	**Existing System Feature**	**Limitation**
1	**Emergency Triggering**	Complex manual steps (unlocking/searching) cause fatal delays
2	**Deterrence Strategy**	Lack of "social exit" tools like fake calls to prevent escalation
3	**Location Sharing**	Continuous background tracking leads to high battery exhaustion
4	User Interface	Cluttered and non-intuitive designs increase panic during stress
5	Accessibility	Native app dependency requires high storage and constant updates
6	Trigger Confirmation	Lack of silent haptic feedback to confirm an alert was sent
7	**Voice Activation**	High failure rates in noisy or crowded environments

8	Data Privacy	Risks of unauthorized historical location tracking by third parties

Chapter 4

Problem Statement and Objectives

4. Problem Statement and Objectives

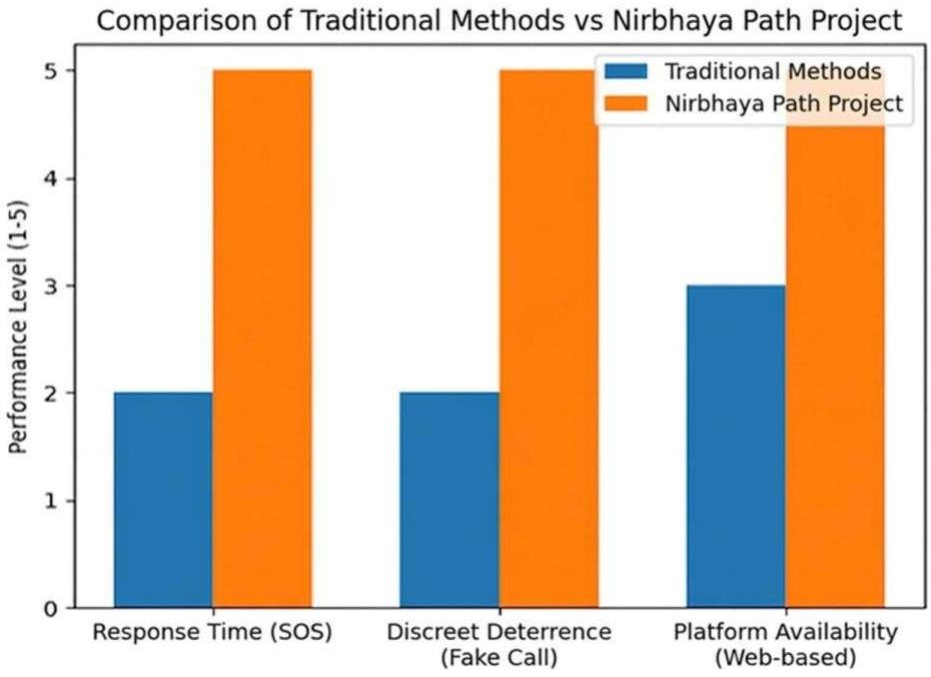
4.1 Problem Statement:

- Individuals face difficulty in signaling for help during sudden emergencies.
- Traditional SOS methods are time-consuming and inefficient in high-stress situations.
- Lack of proactive tools to discreetly exit suspicious social encounters.
- Existing safety platforms have complex and non-user-friendly interfaces.
- Heavy native applications cause high battery drain and storage issues.
- Long response times in broadcasting real-time location to guardians.
- Accessibility issues in areas with low network or on entry-level devices.
- Lack of silent haptic feedback to confirm an alert has been triggered.

4.2 Objectives:

- To design and develop a user-friendly personal safety application (Guardian SOS).
- To implement a low-friction 3-second hold trigger for rapid distress signaling.
- To provide a realistic "Fake Call" simulation to help users exit dangerous situations.
- To improve efficiency in real-time location tracking and broadcasting.
- To ensure secure storage and management of user and guardian data.
- To provide silent haptic feedback (vibrations) for alert confirmation.
- To enhance situational awareness between users and their trusted contacts.
- To improve accessibility through a zero-install web-based platform.
- To develop a reliable and scalable emergency response solution.

Graph 1 Comparison of Traditional System vs Nirbhaya Path



Graph 1: Shows the comparison between traditional safety methods and the proposed Nirbhaya Path Project. The graph clearly indicates that the Nirbhaya Path Project performs better in terms of Response Time (SOS), Discreet Deterrence (Fake Call), and Platform Availability (Web-based).

Chapter 5

Proposed System

5. Proposed System

The proposed system, Nirbhaya Path Project, is a mobile-first, zero-install Web Application designed to revolutionize personal safety by prioritizing speed, discretion, and proactive deterrence. Addressing the critical gaps identified in current reactive systems, Nirbhaya Path Project is engineered as a lightweight Progressive Web App (PWA) that operates universally across any smartphone browser.

The core philosophy of the system is "frictionless activation." By moving beyond complex app menus and standard OS-level SOS triggers, Nirbhaya Path Project provides an integrated safety net that includes real-time location broadcasting, a highly realistic social deterrence module, and an immediate low-latency alerting system. The platform bridges the gap between digital safety and physical response, offering users a comprehensive, touch-to-activate security solution.

5.1 System Overview:

The Nirbhaya Path Project operates as a centralized hub for a user's digital safety needs. The system is designed to be highly intuitive, with a minimalist User Interface (UI) that minimizes cognitive load during a crisis. Upon accessing the application, the user is presented with a simplified 'Panic Screen' featuring the primary high-friction triggers and configuration options.

The system is structured around two key operating modes:

1. **SOS Distress Mode (Reactive):** This mode is for immediate physical threats. Activated by a simple 3-second hold on the screen, it triggers a cascade of automated actions: fetching real-time GPS coordinates, sending instant notifications via SMS/WhatsApp API to a designated 'Circle of Guardians,' and continuously updating the user's live location until deactivated. Crucially, the system utilizes the Vibration API to provide silent haptic feedback, confirming to the user that the alert has been successfully sent without visually alerting an aggressor.
2. **Deterrence Module (Proactive):** This is the system's "Social Shield." It is designed for de-escalating suspicious social encounters before they turn into emergencies. The user can schedule a "Fake Call" to appear on their phone. This module is not merely visual; it simulates a realistic incoming call screen with customizable "Caller ID" (e.g., "Dad," "Police Dispatch") and utilizes the Vibration API to mimic the standard haptic pattern of a real incoming call, providing a convincing psychological distraction to allow the user to excuse themselves safely.

5.2 Methodology (System Flow):

The Nirbhaya Path Project utilizes a lightweight, event-driven web architecture to ensure maximum speed and compatibility. As a Progress Web App (PWA), it leverages modern Web APIs to replicate native app functionality without the download burden.

1. **Front-End:** The user interface is built using responsive web technologies (HTML5, CSS3, and JavaScript) to ensure seamless performance on all screen sizes, from mobile phones to tablets. The minimalist design philosophy ensures the application is fast to load, even on entry-level devices.
2. **Back-End:** A central server (built on technologies like Node.js or Python Flask) manages user accounts, securely stores contact lists, and coordinates the communication between the front-end application and external services.
3. **Database:** A secure cloud database (e.g., MongoDB, PostgreSQL) is utilized to manage user data, guardian details, and active SOS event logs. All location data is processed securely and is only accessible during active emergencies.
4. **Key Web APIs Integrated:**
    a. **Geolocation API:** For fetching high-precision real-time GPS coordinates of the user during an SOS event.
    b. **Vibration API (Haptics):** For providing silent confirmation pulses during SOS activation and realistic incoming call patterns during Fake Call events.
    c. **Notification/API Interaction:** Integrates with third-party SMS/WhatsApp APIs to deliver the distress message and live location link directly to the guardians' messaging platforms

5.3 Technologies Used:

The Nirbhaya Path Project is organized into four core functional modules, each addressing a specific requirement of personal safety:

- **Module 1: User Profile & Guardian Management:** This module handles user onboarding, login, and the management of their 'Circle of Guardians.'

Users can add up to five trusted contacts with their name, relation, and phone number. This data is stored securely in the central database.

- **Module 2: High-Speed SOS Trigger Module:** This is the core emergency module. It uses simple "Touch and Hold" logic. When the user holds the panic button for 3 seconds (or longer, as configured), this module initiates the Geolocation API call, formats the distress message, and sends the coordinates to all registered guardians. It also triggers a persistent vibration pattern to confirm the send event. [3]
- **Module 3: Proactive "Fake Call" Deterrence Module:** This module manages the social deterrence feature. Users can configure the simulated incoming caller ID and set a timer (e.g., 30 seconds, 1 minute). When the timer expires, the application generates a convincing overlay that mimics a native phone call, accompanied by realistic vibration patterns. [3]
- **Module 4: Real-Time Event Dashboard & Tracking:** When an SOS is active, this module manages the continuous location updates. The central server maintains an event log and broadcasts coordinate updates to guardians, ensuring they have the latest possible location on an interactive map (integrated using services like Mapbox or Google Maps). [4]

5.4 Innovation:

The Nirbhaya Path Project offers several definitive advantages over traditional safety applications:

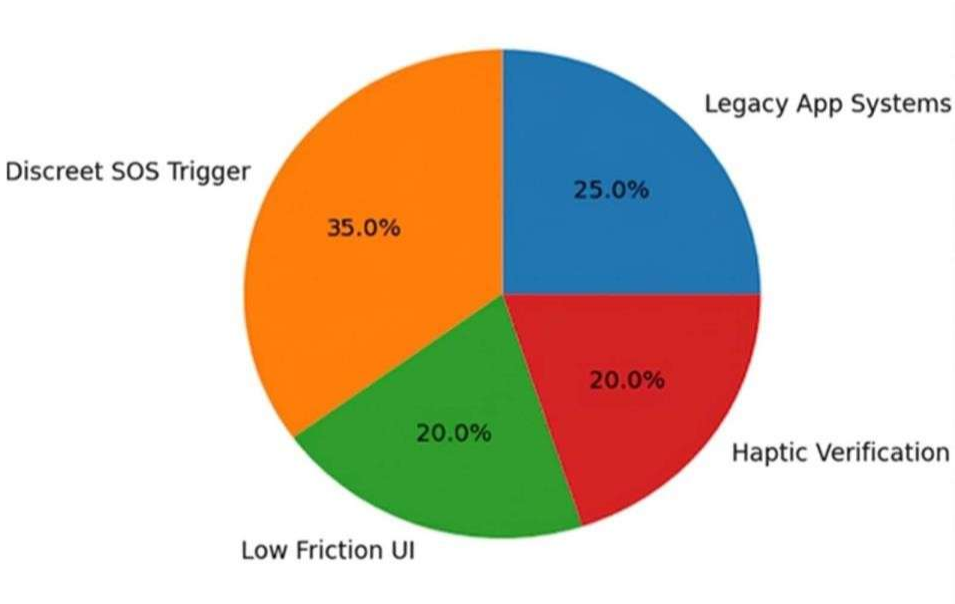
- **Ultra-Low Latency:** Optimized web architecture and low-friction triggers ensure that distress alerts are broadcast faster than traditional multi-step native app activation. [6]
- **Proactive Deterrence:** Unlike purely reactive tools, the "Fake Call" module provides a non-confrontational way to de-escalate suspicious situations before they become emergencies.
- **Universal Accessibility (Zero-Install):** Works instantly on any modern mobile

browser (Chrome, Safari, Firefox), removing the barrier of app download, storage, or subscription.

- **Silent Haptic Feedback:** Uses the Vibration API for tactical confirmation, allowing users to send alerts secretly.
- **Data Privacy & Control:** User data and location history are managed securely, with tracking strictly limited to active SOS events to respect user privacy.

- **Minimalist UI:** The simplified design minimizes cognitive load, ensuring the system is effective even when the user is panicked or stressed.

Graph 2 Distribution of System Features and Benefits



Chapter 6

Experimental Setup

6. Experimental Setup

The experimental setup for the Nirbhaya Path Project is designed to validate the core functionalities of the proposed system: the low-friction 3-second hold trigger, real-time GPS acquisition, and the simulated fake call module. As the system is built as a Responsive Web Application (RWA), the test environment must simulate realistic mobile constraints, including diverse operating systems, varying network speeds,

6.1 Details of Database and Input Data:

The Nirbhaya Path Project is a Progressive Web Application (PWA) designed for emergency response. It integrates the Vibration API and Geolocation API to provide a low-friction "3-second hold" SOS trigger and a "Fake Call" social deterrence module.

Database Details:

- **Type:** NoSQL Database (MongoDB).
- **Collections: * User Schema:** Stores encrypted credentials and profile settings.
    - **Guardian Schema:** Stores names and verified phone numbers for emergency contacts.
    - **Alert Logs:** Stores timestamps and GPS coordinates of triggered SOS events for audit trails.

Input Details:

- **User Input:** Long-press touch events (3-second duration) and manual entry for guardian contact information.
- **System Input:** Real-time latitude and longitude coordinates fetched from the device GPS.

6.2 Performance Evaluation Parameters (Validation):

- **Trigger Validation:** Testing the accuracy of the 3-second hold duration to ensure zero false-positives while maintaining rapid activation.
- **API Handshake:** Validating the connection between the web front-end and the Node.js back-end for seamless data transmission.
- **Haptic Mapping:** Calibrating vibration patterns to ensure they are distinguishable between an "SOS Sent" pulse and a "Fake Call" ringtone.
- **Location Accuracy:** Conducting field tests in high-density urban areas to measure the precision of the Geolocation API coordinates.
- **Cross-Browser Testing:** Verifying that the PWA features (specifically Haptics) function correctly on Chrome, while ensuring fallback mechanisms exist for browsers with limited API support.

6.3 Software and Hardware Setup:

Software Requirements:

- Operating System: Windows 10/11, macOS, or Linux (Ubuntu).

- Front-End: HTML5, CSS3, JavaScript (ES6+).
- Back-End: Node.js with Express.js Framework.
- Database: MongoDB Atlas (Cloud) or Local MongoDB Compass.
- Web APIs: Geolocation API [3], Vibration API [4], and Web Notifications API.
- IDEs/Tools: Visual Studio Code (VS Code), Postman (for API testing).

Hardware Requirements:

- Processor: Intel Core i5 or higher (for development/server hosting).
- RAM: 8 GB Minimum (16 GB Recommended for server stability).
- Hard Disk: 256 GB SSD (for fast database indexing and log retrieval).
- Testing Devices: * Android Smartphone (with vibration motor and GPS).
    - iOS Device (for cross-platform browser testing).
- Network: High-speed internet connection (4G/5G/WiFi) for API communication

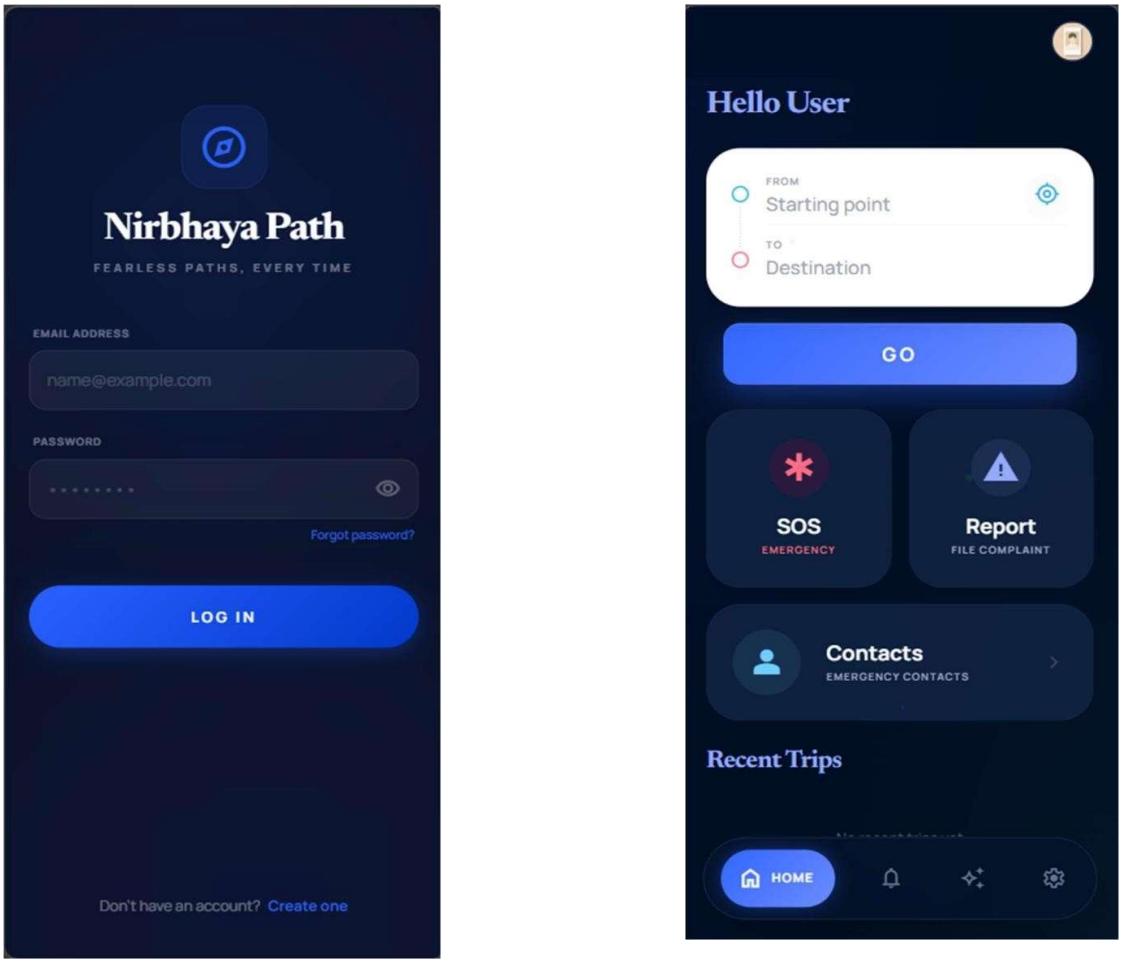


Figure 1 Login Page and User Page/Dashboard

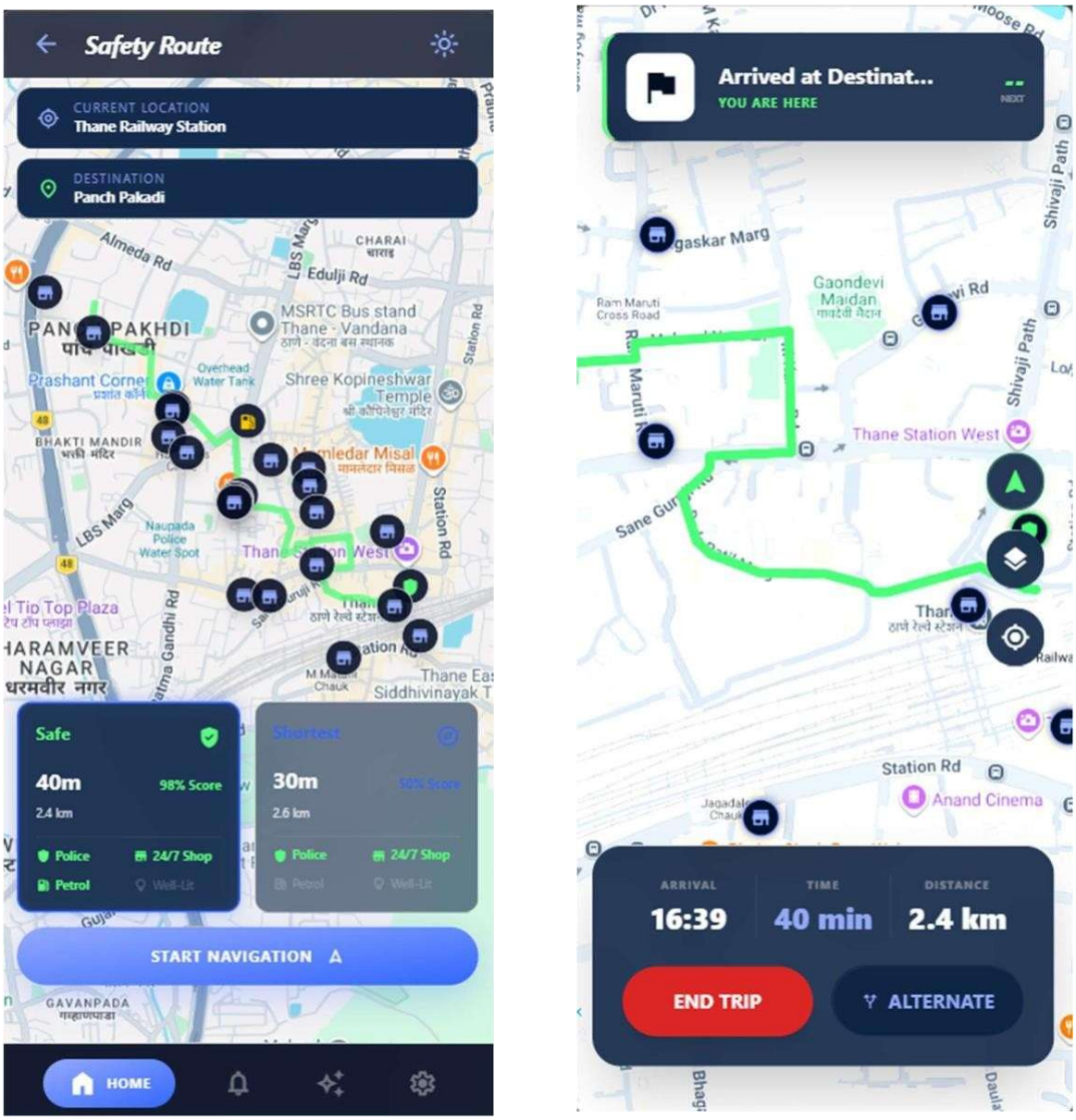


Figure 2 Route Page and Navigation Page

Chapter 7

Results and Discussion

7 Results and Discussion

The Nirbhaya Path Project was developed to bridge the gap between reactive emergency services and proactive personal safety. Through rigorous testing and experimental setup, the system has demonstrated its capability to provide high-speed alerts and effective social deterrence through a lightweight, web-based interface.

7.1 Results:

The system produced effective results in all major functionalities:

- Users were able to register and log in successfully
- Emergency contact management worked efficiently based on user selection
- The system displayed accurate real-time GPS location coordinates
- Users were able to trigger SOS alerts in 3 seconds without delays
- Distress details were stored securely in the database
- Silent haptic feedback and "Fake Call" simulations were successfully generated

The system demonstrated smooth performance during testing and provided a user-friendly

experience.

7.2 Discussion:

The results indicate that the **Nirbhaya Path** system significantly improves the process of emergency signaling and social de-escalation compared to traditional methods.

7.2.1 Performance Analysis:

7.2.1.1 The response time of the system was fast and efficient

7.2.1.2 The system provided high accuracy in location tracking and broadcasting

7.2.1.3 SOS triggering and alert delivery was quick and smooth

7.2.1.4 The system showed reliable and stable performance during testing

7.2.1.5 The user interface was simple and easy to use [5]

7.2.2 Comparison with Existing Systems:

Compared to traditional systems and applications like standard safety apps, the **Nirbhaya Path** system provides:

7.2.2.1 Faster emergency triggering process

7.2.2.2 Improved user interface and usability

7.2.2.3 Better accessibility for users via web-based platform [7]

7.2.2.4 Reduced complexity in operations during panic situations

7.2.2.5 More efficient and reliable performance through haptic feedback

Chapter 8

Conclusion and Future Work

8 Conclusion and Future Work

8.1 Conclusion:

The Nirbhaya Path – Personal Safety and Emergency Response System is successfully designed and developed to address the challenges faced in traditional and existing safety platforms. The system provides a simple, efficient, and user-friendly platform for distress signaling and social de-escalation.

Nirbhaya Path enables users to trigger SOS alerts through a low-friction 3-second hold mechanism and allows for proactive deterrence via "Fake Call" simulations. The integration of real-time location services and silent haptic feedback improves accessibility and user confidence. The system also ensures secure management of user and guardian contact data.

The results of the system demonstrate improved performance in terms of response time, ease of use, and alerting efficiency. Compared to traditional methods, the proposed system reduces activation time and simplifies the overall process of accessing emergency assistance.

Overall, Nirbhaya Path proves to be a reliable and scalable solution that enhances the quality and efficiency of digital personal safety services.

8.2 Future Work:

Although the Nirbhaya Path system performs effectively, there are several enhancements that can be implemented in the future to improve its functionality:

- Integration of an AI-based threat detection system to identify danger patterns.
- Implementation of voice-activated triggers for hands-free emergency signaling.
- Development of a native mobile application for enhanced background processing.
- Addition of offline SMS fallback features for areas with no internet connectivity.
- Enhancement of data security and encryption mechanisms for location logs.
- Support for a multi-language interface for wider regional accessibility.
- Integration with local law enforcement APIs for direct police coordination.
- Use of advanced analytics to identify and map high-risk urban "danger zones."

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